Cornstarch Goo

01 Why does a fast smack stop at the top?



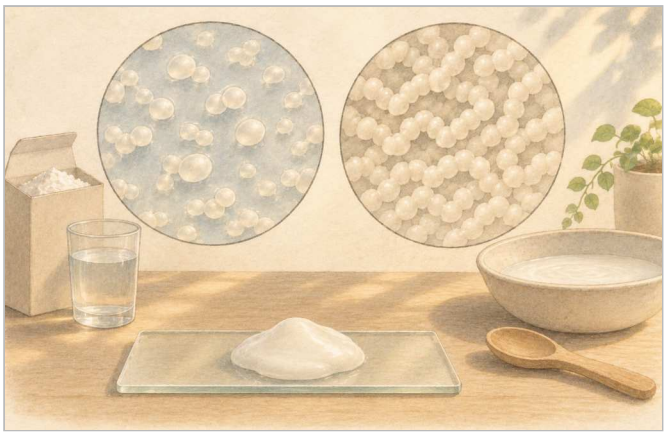
Slow sinks deep; a fast smack holds.

Look closer. Compare the deep, sunk spoon with the dull dent in the other bowl.

A little explanation. Most liquids, like water, stay just as runny however fast you push. Isaac Newton described that rule, so this goo is called non-Newtonian. A slow push lets it flow. A fast smack makes it act like a solid for a moment.

Try this. Mix about 2 scoops of cornstarch with 1 scoop of water. Rest a spoon on top, then tap the goo fast with another. Which one sinks?

02 What happens to the grains when you push fast?



Slow pushes let grains slide. Fast pushes jam them.

Look closer. Find the grains floating apart and the grains crowded into chains.

A little explanation. Each cornstarch grain is so tiny that about five fit across one hair. Water sits between them like a slippery layer. A slow push lets the water slip aside. A fast push shoves the grains together before the water can move, so they lock up. Scientists call this jamming.

Try this. Drag a spoon slowly through the goo, then quickly. Feel how much harder your arm has to work.

03 Why does squeezed goo look dry?



Squeezed goo looks dry, then melts.

Look closer. Look at the dull, cracked skin of the ball and the shiny puddle beside it.

A little explanation. Jammed grains need a little more room, so they poke up through the water at the surface. The skin turns dull and cracks. Stop squeezing, and the water covers them again, so the goo shines and melts.

Try this. Roll a lump of goo between your palms, then stop and open your hands. Watch it melt and drip.

04 Why does wet sand go pale around a foot?



A step makes wet sand look dry for a moment.

Look closer. Find the pale ring around the press mark in the wet sand.

A little explanation. A step pushes the sand grains a little apart. Water sinks into the new gaps, so the top looks dry for a moment. Osborne Reynolds explained this in 1885 and named it dilatancy. Cornstarch goo does a quick version of the same trick.

Try this. Next time you are at a beach, step on firm wet sand. Watch the pale ring appear, then lift your foot.